Active Maintenance Report: green_etal_2016

Green, Krasno, Coppock, Farrer, Lenoir and Zingher (2016, Electoral Studies): The Effects of Lawn Signs on Vote Outcomes

2026-08-01

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This repository holds the actively maintained replication code for Green, Krasno, Coppock, Farrer, Lenoir and Zingher (2016), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1016/j.electstud.2015.12.002
Replication archive	10.7910/DVN/K2TLDB
Pre-analysis plan (Experiment 3)	osf.io/umysq
Pre-analysis plan (Experiment 4)	osf.io/ejfgk

The data are not redistributed here. The deposit is 12 MB across 33 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `green_etal_2016.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 33 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, metafor, gridExtra, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about two minutes, almost all of it in the randomization inference, which refits 120,000 pairs of weighted models across 40,000 permutations. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes. All six analysis scripts execute without error on a current R installation, with no hardcoded paths, no functions called before they are defined, and no deprecated calls that have since been removed. Every package they name still installs from CRAN, including `beepr` 2.0, which only one of the random-assignment scripts calls, to play a sound on completion.

Running is not the same as reproducing, and one number changes. `Experiment_2_Analysis.R` picks the observed treatment assignment out of a matrix of 10,000 admissible randomizations with `set.seed(12345); sample(1:10000, 1)`. R 3.6.0 changed how `sample()` converts uniform draws to integers, so that line returned column 7210 when the paper was written and returns column 8243 today. Column 7210 is the assignment that was actually deployed, and it is the one the archive's own pre-saved `exp_2.RData` encodes, so the archive contains the right answer and the live script path no longer finds it. 10 of the 146 claims an archive script can be checked against therefore fail to reproduce, all of them in Table 4 and in the Experiment 2 randomization inference p-value, and all from that one line.

1.2 Does the maintained rewrite reproduce the paper?

Yes, without exception. All 191 verifiable ground truth claims match the published values to reported precision, including every panel label of Figure 2. The remaining 8 recorded quantities are randomization inference p-values for vote margin and turnout, which the scripts compute but the article and its appendix never state; they are marked unverifiable rather than matched.

The rewrite reaches Experiment 2 through the pre-saved `exp_2.RData` object rather than through the seed, which is what makes it immune to the R version change. Reaching it any other way is a live hazard rather than a hypothetical one: until August 2026 the rewrite's randomization inference script took Experiment 2's observed assignment from its own cleaning script, which redraws the seed, and consequently reported 0.79 where the article reports 0.90. That script now reads the same canonical object as everything else.

2 Paper overview

Citation: Green, D. P., Krasno, J. S., Coppock, A., Farrer, B. D., Lenoir, B. and Zingher, J. N. (2016). "The effects of lawn signs on vote outcomes: Results from four randomized field experiments." *Electoral Studies*, 41, 143-150. DOI: 10.1016/j.electstud.2015.12.002

Summary: Four randomized field experiments test whether lawn signs change vote outcomes. In each, geographic units (electoral districts or precincts) were assigned to receive signs, to be adjacent to a unit that received signs, or to neither, so that each experiment estimates a direct effect and a spillover effect at once. Because adjacency makes the assignment probabilities unequal across units, every estimate is inverse-probability weighted, and inference is randomization based: an observed F-statistic for the joint null of no direct and no indirect effect is compared against 10,000 pre-computed admissible randomizations. The four experiments are a 2012 congressional general election in New York, a 2013 mayoral primary in Albany, a 2013 gubernatorial general election in Virginia, and a 2014 state senate primary in Pennsylvania. Pooling the four by fixed-effects meta-analysis gives a direct effect of 1.7 percentage points of vote share (SE 0.7) and a spillover effect of 1.5 points (SE 0.6).

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
Lawn_Signs_Source.R	Clean (sourced by all scripts)	No changes required
Experiment_1_Analysis.R	Clean	No changes required
Experiment_2_Analysis.R	Runs clean, but Table 4 and the Experiment 2 p-value no longer reproduce	set.seed(12345, sample.kind = 'Rounding'), or read the deposited exp_2.RData
Experiment_3_Analysis.R	Clean	No changes required
Experiment_4_Analysis.R	Clean	No changes required
Pooled_analysis.R	Clean	No changes required
in_text_calculations.R	Clean	No changes required
Experiment_1_RA_Function.R	Parses; regenerates a deposited matrix, not re-run	None needed: its output is deposited
Experiment_2_RA_Function.R	Parses; regenerates a deposited matrix, not re-run	None needed: its output is deposited
Experiment_3_RA_Function.R	Parses; regenerates a deposited matrix, not re-run	None needed: its output is deposited
Experiment_4_RA_Function.R	Parses; needs beepr, which installs from CRAN	install.packages('beepr')

Every package the archive names still installs from CRAN: `stargazer`, `xtable`, `sandwich`, `lmtest`, `ggplot2`, `dplyr`, `randomizr`, `beepr` and `rmeta`. The last of these is the fragile one. `rmeta` 3.0 has not been updated since March 2018 and survives on CRAN by its archiving policy rather than by maintenance, and `Pooled_analysis.R` depends on it for a single call to `meta.summaries()`. The maintained rewrite pools with `metafor::rma(method = "FE")` instead, so it does not inherit the exposure.

The four `*_RA_Function.R` scripts regenerate the permutation matrices from scratch. They parse cleanly and their packages install, but they were not re-run to completion: they are stochastic, they take a long time, and the matrices they build are themselves deposited, so re-running them would replace a deposited object with a different one rather than check anything.

3.1 The Experiment 2 seed

The one substantive failure has nothing to do with packages. `Experiment_2_Analysis.R` opens by drawing the observed treatment assignment out of the deposited permutation matrix:

```
set.seed(12345)
chosen.rand <- sample(1:10000, 1)
```

R 3.6.0, released in April 2019, replaced the "Rounding" method that `sample()` used to convert uniform variates to integers with a "Rejection" method. The same seed therefore selects a different column than it did when the analysis was run: 7210 under the old method, 8243 under the new one. The two columns are different treatment assignments and give different answers.

Column 7210 is the assignment the experiment actually used. Three independent checks agree on that: it is the column the archive's own pre-saved `exp_2.RData` encodes, it is the column whose treated set matches the authors' record of which districts received signs, and it is what `set.seed(12345, sample.kind = "Rounding")` returns on any R version. Column 7210 also reproduces the published Table 4 and the published Experiment 2 p-value; column 8243 reproduces neither.

Table 2: What the Experiment 2 permutation column decides.

Quantity	Published	Column 7210 (R $<$ 3.6)	Column 8243 (R 3.6+)
Table 4 M1 treated	0.009	0.009	0.130
Table 4 M2 treated	-0.014	-0.014	0.019
Section 5.2 p-value	0.90	0.897	0.794

The archive is not wrong. It contains both the script and the object the script was meant to produce, and the object is correct. What it lacks is any signal that the two have come apart, which is the failure this maintenance program exists to catch: a reader running the deposited script today gets numbers that differ from the paper, with no error and no warning.

4 Number-by-number comparison

Table 3: Ground truth: published value against the value the deposited scripts produce on current R. A blank Match means the quantity is not stated in the article or its appendix, or that the archive prints no comparable value.

Location	Exp.	Arm	Model	Quantity	Paper	Archive	Match
table_2	all	Control	n	N	16	16	1
table_2	all	Adjacent	n	N	49	49	1

Table 3: Ground truth: published value against the value the deposited scripts produce on current R. A blank Match means the quantity is not stated in the article or its appendix, or that the archive prints no comparable value. *(continued)*

Location	Exp.	Arm	Model	Quantity	Paper	Archive	Match
table_2	all	Treated	n	N	23	23	1
table_2	all	Control	n	N	13	13	1
table_2	all	Adjacent	n	N	41	41	1
table_2	all	Treated	n	N	15	15	1
table_2	all	Control	n	N	25	25	1
table_2	all	Adjacent	n	N	76	76	1
table_2	all	Treated	n	N	30	30	1
table_2	all	Control	n	N	24	24	1
table_2	all	Adjacent	n	N	44	44	1
table_2	all	Treated	n	N	20	20	1
table_3	1	Treated	M1	coef	0.025	0.025	1
table_3	1	Treated	M1	SE	0.027	0.027	1
table_3	1	Adjacent	M1	coef	0.037	0.037	1
table_3	1	Adjacent	M1	SE	0.027	0.027	1
table_3	1	Treated	M2	coef	0.025	0.025	1
table_3	1	Treated	M2	SE	0.017	0.017	1
table_3	1	Adjacent	M2	coef	0.018	0.018	1
table_3	1	Adjacent	M2	SE	0.016	0.016	1
table_3	1	all	M1	N	88	88	1
table_3	1	all	M1	R2	0.031	0.031	1
table_3	1	all	M2	R2	0.823	0.823	1
table_4	2	Treated	M1	coef	0.009	0.13	0
table_4	2	Treated	M1	SE	0.054	0.08	0
table_4	2	Adjacent	M1	coef	0.012	0.137	0
table_4	2	Adjacent	M1	SE	0.046	0.072	0
table_4	2	Treated	M2	coef	-0.014	0.019	0
table_4	2	Treated	M2	SE	0.057	0.053	0
table_4	2	Adjacent	M2	coef	0.004	0.028	0
table_4	2	Adjacent	M2	SE	0.045	0.035	0
table_4	2	all	M1	N	69	69	1
table_4	2	all	M1	R2	0.001	0.13	0
table_5	3	Treated	M1	coef	0.042	0.042	1
table_5	3	Treated	M1	SE	0.016	0.016	1
table_5	3	Adjacent	M1	coef	0.042	0.042	1
table_5	3	Adjacent	M1	SE	0.013	0.013	1
table_5	3	Treated	M2	coef	0.018	0.018	1
table_5	3	Treated	M2	SE	0.009	0.009	1
table_5	3	Adjacent	M2	coef	0.018	0.018	1
table_5	3	Adjacent	M2	SE	0.007	0.007	1
table_5	3	all	M1	N	131	131	1
table_5	3	all	M1	R2	0.094	0.094	1
table_5	3	all	M2	R2	0.825	0.825	1
table_6	4	Treated	M1	coef	-0.013	-0.013	1
table_6	4	Treated	M1	SE	0.028	0.028	1
table_6	4	Adjacent	M1	coef	-0.023	-0.023	1
table_6	4	Adjacent	M1	SE	0.022	0.022	1
table_6	4	Treated	M2	coef	-0.012	-0.012	1
table_6	4	Treated	M2	SE	0.026	0.026	1
table_6	4	Adjacent	M2	coef	-0.02	-0.02	1
table_6	4	Adjacent	M2	SE	0.021	0.021	1
table_6	4	all	M1	N	88	88	1
table_6	4	all	M1	R2	0.012	0.012	1
table_6	4	all	M2	R2	0.172	0.172	1
table_7	all	Treated	pooled	coef_exp1	0.025	0.025	1
table_7	all	Treated	pooled	SE_exp1	0.017	0.017	1
table_7	all	Treated	pooled	coef_exp2	-0.014	-0.014	1
table_7	all	Treated	pooled	SE_exp2	0.057	0.057	1
table_7	all	Treated	pooled	coef_exp3	0.018	0.018	1
table_7	all	Treated	pooled	SE_exp3	0.009	0.009	1
table_7	all	Treated	pooled	coef_exp4	-0.012	-0.012	1
table_7	all	Treated	pooled	SE_exp4	0.026	0.026	1
table_7	all	Treated	pooled	coef_pooled	0.017	0.017	1
table_7	all	Treated	pooled	SE_pooled	0.007	0.007	1
table_7	all	Adjacent	pooled	coef_exp1	0.018	0.018	1

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Location	Exp.	Arm	Model	Quantity	Paper	Archive	Match
table_7	all	Adjacent	pooled	SE_exp1	0.016	0.016	1
table_7	all	Adjacent	pooled	coef_exp2	0.004	0.004	1
table_7	all	Adjacent	pooled	SE_exp2	0.045	0.045	1
table_7	all	Adjacent	pooled	coef_exp3	0.018	0.018	1
table_7	all	Adjacent	pooled	SE_exp3	0.007	0.007	1
table_7	all	Adjacent	pooled	coef_exp4	-0.02	-0.02	1
table_7	all	Adjacent	pooled	SE_exp4	0.021	0.021	1
table_7	all	Adjacent	pooled	coef_pooled	0.015	0.015	1
table_7	all	Adjacent	pooled	SE_pooled	0.006	0.006	1
table_8	1	Treated	het	coef	0.031	0.031	1
table_8	1	Treated	het	SE	0.017	0.017	1
table_8	1	Adjacent	het	coef	0.021	0.021	1
table_8	1	Adjacent	het	SE	0.014	0.014	1
table_8	1	party_share	het	coef	0.027	0.027	1
table_8	1	party_share	het	SE	0.042	0.042	1
table_8	1	Treated:party_share	het	coef	0.031	0.031	1
table_8	1	Treated:party_share	het	SE	0.03	0.03	1
table_8	1	Adjacent:party_share	het	coef	0.011	0.011	1
table_8	1	Adjacent:party_share	het	SE	0.025	0.025	1
table_8	1	all	het	N	88	88	1
table_8	1	all	het	R2	0.829	0.829	1
table_8	2	Treated	het	coef	-0.009	-0.009	1
table_8	2	Treated	het	SE	0.06	0.06	1
table_8	2	Adjacent	het	coef	0.011	0.011	1
table_8	2	Adjacent	het	SE	0.046	0.046	1
table_8	2	party_share	het	coef	-0.02	-0.02	1
table_8	2	party_share	het	SE	0.033	0.033	1
table_8	2	Treated:party_share	het	coef	0.033	0.033	1
table_8	2	Treated:party_share	het	SE	0.063	0.063	1
table_8	2	Adjacent:party_share	het	coef	0.053	0.053	1
table_8	2	Adjacent:party_share	het	SE	0.033	0.033	1
table_8	2	all	het	N	69	69	1
table_8	2	all	het	R2	0.277	0.277	1
table_8	3	Treated	het	coef	0.02	0.02	1
table_8	3	Treated	het	SE	0.009	0.009	1
table_8	3	Adjacent	het	coef	0.02	0.02	1
table_8	3	Adjacent	het	SE	0.007	0.007	1
table_8	3	party_share	het	coef	0.028	0.028	1
table_8	3	party_share	het	SE	0.009	0.009	1
table_8	3	Treated:party_share	het	coef	0.01	0.01	1
table_8	3	Treated:party_share	het	SE	0.008	0.008	1
table_8	3	Adjacent:party_share	het	coef	0.008	0.008	1
table_8	3	Adjacent:party_share	het	SE	0.008	0.008	1
table_8	3	all	het	N	131	131	1
table_8	3	all	het	R2	0.829	0.829	1
table_8	4	Treated	het	coef	-0.014	-0.014	1
table_8	4	Treated	het	SE	0.026	0.026	1
table_8	4	Adjacent	het	coef	-0.02	-0.02	1
table_8	4	Adjacent	het	SE	0.021	0.021	1
table_8	4	party_share	het	coef	0.02	0.02	1
table_8	4	party_share	het	SE	0.021	0.021	1
table_8	4	Treated:party_share	het	coef	-0.021	-0.021	1
table_8	4	Treated:party_share	het	SE	0.027	0.027	1
table_8	4	Adjacent:party_share	het	coef	-0.016	-0.016	1
table_8	4	Adjacent:party_share	het	SE	0.02	0.02	1
table_8	4	all	het	N	88	88	1
table_8	4	all	het	R2	0.181	0.181	1
ri_table	1	all	RI	p_margin		0.16	
ri_table	1	all	RI	p_share	0.22	0.22	1
ri_table	1	all	RI	p_turnout		0.53	
ri_table	2	all	RI	p_margin		0.29	
ri_table	2	all	RI	p_share	0.9	0.79	0
ri_table	2	all	RI	p_turnout		0.44	
ri_table	3	all	RI	p_margin		0.26	

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Location	Exp.	Arm	Model	Quantity	Paper	Archive	Match
ri_table	3	all	RI	p_share	0.02	0.02	1
ri_table	3	all	RI	p_turnout		0.33	
ri_table	4	all	RI	p_margin		0.88	
ri_table	4	all	RI	p_share	0.77	0.77	1
ri_table	4	all	RI	p_turnout		0.85	
appendix_C1	1	Treated	M1	coef	15.582	15.582	1
appendix_C1	1	Treated	M1	SE	28.327	28.327	1
appendix_C1	1	Adjacent	M1	coef	-9.242	-9.242	1
appendix_C1	1	Adjacent	M1	SE	27.305	27.305	1
appendix_C1	1	Treated	M2	coef	34.786	34.786	1
appendix_C1	1	Treated	M2	SE	18.207	18.207	1
appendix_C1	1	Adjacent	M2	coef	11.713	11.713	1
appendix_C1	1	Adjacent	M2	SE	17.885	17.885	1
appendix_C1	1	Treated	M1_turnout	coef	53.002	53.002	1
appendix_C1	1	Treated	M1_turnout	SE	48.748	48.748	1
appendix_C1	1	Adjacent	M1_turnout	coef	110.329	110.329	1
appendix_C1	1	Adjacent	M1_turnout	SE	52.796	52.796	1
appendix_C1	1	Treated	M2_turnout	coef	10.331	10.331	1
appendix_C1	1	Treated	M2_turnout	SE	15.741	15.741	1
appendix_C1	1	Adjacent	M2_turnout	coef	11.914	11.914	1
appendix_C1	1	Adjacent	M2_turnout	SE	11.759	11.759	1
footnote_4	all	all	intext	total_turnout	241613	241613	1
footnote_4	all	all	intext	cost_per_vote_direct	3.18	3.18	1
footnote_4	all	all	intext	cost_per_vote_total	1.69	1.69	1
figure_2	Agnostic	Prior		posterior_mean	0		
figure_2	Agnostic	Prior		posterior_sd	0.05		
figure_2	Agnostic	Prior		pr_ate_positive	0.5		
figure_2	Agnostic	Update 1		posterior_mean	0.022		
figure_2	Agnostic	Update 1		posterior_sd	0.016		
figure_2	Agnostic	Update 1		pr_ate_positive	0.915		
figure_2	Agnostic	Update 2		posterior_mean	0.019		
figure_2	Agnostic	Update 2		posterior_sd	0.016		
figure_2	Agnostic	Update 2		pr_ate_positive	0.895		
figure_2	Agnostic	Update 3		posterior_mean	0.019		
figure_2	Agnostic	Update 3		posterior_sd	0.008		
figure_2	Agnostic	Update 3		pr_ate_positive	0.993		
figure_2	Agnostic	Update 4		posterior_mean	0.016		
figure_2	Agnostic	Update 4		posterior_sd	0.007		
figure_2	Agnostic	Update 4		pr_ate_positive	0.988		
figure_2	Optimist	Prior		posterior_mean	0.05		
figure_2	Optimist	Prior		posterior_sd	0.05		
figure_2	Optimist	Prior		pr_ate_positive	0.841		
figure_2	Optimist	Update 1		posterior_mean	0.027		
figure_2	Optimist	Update 1		posterior_sd	0.016		
figure_2	Optimist	Update 1		pr_ate_positive	0.955		
figure_2	Optimist	Update 2		posterior_mean	0.024		
figure_2	Optimist	Update 2		posterior_sd	0.016		
figure_2	Optimist	Update 2		pr_ate_positive	0.941		
figure_2	Optimist	Update 3		posterior_mean	0.02		
figure_2	Optimist	Update 3		posterior_sd	0.008		
figure_2	Optimist	Update 3		pr_ate_positive	0.996		
figure_2	Optimist	Update 4		posterior_mean	0.017		
figure_2	Optimist	Update 4		posterior_sd	0.007		
figure_2	Optimist	Update 4		pr_ate_positive	0.991		
figure_2	Skeptic	Prior		posterior_mean	0		
figure_2	Skeptic	Prior		posterior_sd	0.01		
figure_2	Skeptic	Prior		pr_ate_positive	0.5		
figure_2	Skeptic	Update 1		posterior_mean	0.006		
figure_2	Skeptic	Update 1		posterior_sd	0.009		
figure_2	Skeptic	Update 1		pr_ate_positive	0.768		
figure_2	Skeptic	Update 2		posterior_mean	0.006		
figure_2	Skeptic	Update 2		posterior_sd	0.009		
figure_2	Skeptic	Update 2		pr_ate_positive	0.754		
figure_2	Skeptic	Update 3		posterior_mean	0.012		

Table 3: Ground truth: published value against the value the deposited scripts produce on current R. A blank Match means the quantity is not stated in the article or its appendix, or that the archive prints no comparable value. *(continued)*

Location	Exp.	Arm	Model	Quantity	Paper	Archive	Match
figure_2	Skeptic	Update 3		posterior_sd	0.006		
figure_2	Skeptic	Update 3		pr_ate_positive	0.977		
figure_2	Skeptic	Update 4		posterior_mean	0.011		
figure_2	Skeptic	Update 4		posterior_sd	0.006		
figure_2	Skeptic	Update 4		pr_ate_positive	0.966		

Of the 199 recorded claims, 146 can be compared against what a deposited script prints. 136 of those match the published value and 10 do not, every one of the failures traceable to the Experiment 2 seed.

5 Maintained rewrite

The rewrite lives in `maintained/`: fifteen scripts covering four cleaning steps, seven published tables, the appendix margin and turnout tables, Figure 2, and the randomization inference. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used.

5.1 Architecture

Each experiment's analysis reads the archive's canonical `exp_N.RData` object, which carries both the analysis frame and that experiment's 10,000-column permutation matrix. The `clean_expN.R` scripts rebuild the same frames from the raw deposited files in tidy form and are the readable account of how each variable is constructed, but no published number is taken from them. That separation is deliberate and load-bearing: `clean_exp2.R` redraws the observed assignment with the same seed the archive uses, and therefore selects the wrong column on any current R.

Weighted models use `estimatr::lm_robust(se_type = "HC2")` in place of `lm()` followed by a `coeftest(fit, vcovHC(fit, type = "HC2"))` wrapper, which is the same estimator with the same standard errors. Pooling uses `metafor::rma(method = "FE")` in place of `rmeta::meta.summaries()`. The randomization inference reimplements the archive's `get_f()` as `f_treat()` in `helpers.R` and loops over the deposited permutation matrices, computing all four experiments rather than carrying any p-value forward as a constant.

5.2 Deprecated patterns replaced

Table 4: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd("")‘	‘here::here()‘
‘lm()‘ + ‘coeftest(vcovHC, 'HC2')‘	‘estimatr::lm_robust(se_type = "HC2")‘
‘rmeta::meta.summaries(method = 'fixed')‘	‘metafor::rma(method = "FE")‘
‘stargazer‘ + ‘sink()‘	‘write_csv()‘ to ‘output/‘
‘xtable‘ + ‘print.xtable‘	‘write_csv()‘ to ‘output/‘
‘within()‘ blocks over base indexing	‘mutate()‘ paragraphs
‘merge()‘	‘left_join()‘
‘multiplot()‘ (grid-based)	‘gridExtra::arrangeGrob()‘
‘pdf()‘ / ‘png()‘ devices	‘ggsave()‘
‘library(beepr)‘ + ‘beep()‘	(omitted)
magrittr pipe	native pipe

6 Randomization inference

The article reports one randomization inference p-value per experiment, for vote share, in sections 5.1 through 5.4. All four reproduce.

Table 5: Randomization inference p-values for the joint null of no direct and no indirect effect, 10,000 permutations each. The article states the vote share p-values only.

Experiment	Outcome	Rewrite	Published
Experiment 1	share	0.2225	0.22
Experiment 1	margin	0.1573	
Experiment 1	turnout	0.5261	
Experiment 2	share	0.8974	0.90
Experiment 2	margin	0.8928	
Experiment 2	turnout	0.6967	
Experiment 3	share	0.0208	0.02
Experiment 3	margin	0.2633	
Experiment 3	turnout	0.3322	
Experiment 4	share	0.7714	0.77
Experiment 4	margin	0.8770	
Experiment 4	turnout	0.8509	

The margin and turnout p-values are computed here for completeness and appear nowhere in the article or its appendix, so the ground truth marks them unverifiable rather than matched. Producing them is cheap: all twelve p-values together, 40,000 permutations and 120,000 pairs of weighted models, take under two minutes.

Every p-value here is computed rather than carried. The permutation matrices are already inside the `exp_3.RData` and `exp_4.RData` objects the table scripts load, so no additional file is read, and the two loops together run in under a minute. A number typed into a script is not a number the script produces.

7 Figure 2 verification

Figure 2 traces how an agnostic, an optimistic and a skeptical observer update a prior about the direct effect as the four experiments arrive, and each of its fifteen panels is labelled with the posterior mean, the posterior standard deviation and the posterior probability that the effect is positive. Those forty-five numbers are the figure's content, and all forty-five reproduce.

Table 6: Figure 2 panel labels, transcribed from page 149 of the article, against the maintained rewrite.

Observer	Panel	Quantity	Paper	Rewrite	Match
	Prior	mean	0.000	0.000	1
	Prior	sd	0.050	0.050	1
	Prior	pr(ATE > 0)	0.500	0.500	1
	Update 1	mean	0.022	0.022	1
	Update 1	sd	0.016	0.016	1
	Update 1	pr(ATE > 0)	0.915	0.915	1
	Update 2	mean	0.019	0.019	1
Agnostic	Update 2	sd	0.016	0.016	1
	Update 2	pr(ATE > 0)	0.895	0.895	1
	Update 3	mean	0.019	0.019	1
	Update 3	sd	0.008	0.008	1
	Update 3	pr(ATE > 0)	0.993	0.993	1
	Update 4	mean	0.016	0.016	1
	Update 4	sd	0.007	0.007	1
	Update 4	pr(ATE > 0)	0.988	0.988	1
	Prior	mean	0.050	0.050	1
	Prior	sd	0.050	0.050	1
	Prior	pr(ATE > 0)	0.841	0.841	1
	Update 1	mean	0.027	0.027	1
	Update 1	sd	0.016	0.016	1
	Update 1	pr(ATE > 0)	0.955	0.955	1
	Update 2	mean	0.024	0.024	1
Optimist	Update 2	sd	0.016	0.016	1
	Update 2	pr(ATE > 0)	0.941	0.941	1
	Update 3	mean	0.020	0.020	1
	Update 3	sd	0.008	0.008	1
	Update 3	pr(ATE > 0)	0.996	0.996	1
	Update 4	mean	0.017	0.017	1
	Update 4	sd	0.007	0.007	1
	Update 4	pr(ATE > 0)	0.991	0.991	1

Table 6: Figure 2 panel labels, transcribed from page 149 of the article, against the maintained rewrite. *(continued)*

Observer	Panel	Quantity	Paper	Rewrite	Match
Skeptic	Prior	mean	0.000	0.000	1
	Prior	sd	0.010	0.010	1
	Prior	pr(ATE $>$ 0)	0.500	0.500	1
	Update 1	mean	0.006	0.006	1
	Update 1	sd	0.009	0.009	1
	Update 1	pr(ATE $>$ 0)	0.768	0.768	1
	Update 2	mean	0.006	0.006	1
	Update 2	sd	0.009	0.009	1
	Update 2	pr(ATE $>$ 0)	0.754	0.754	1
	Update 3	mean	0.012	0.012	1
	Update 3	sd	0.006	0.006	1
	Update 3	pr(ATE $>$ 0)	0.977	0.977	1
	Update 4	mean	0.011	0.011	1
	Update 4	sd	0.006	0.006	1
	Update 4	pr(ATE $>$ 0)	0.966	0.966	1

The rewrite updates on the fitted coefficients at full precision, reading them from `maintained/output/table_7_pooled.csv`, which is what the original does. Retyping the four direct effects at the three decimal places Table 7 prints would carry a rounding into a calculation rather than into a display.

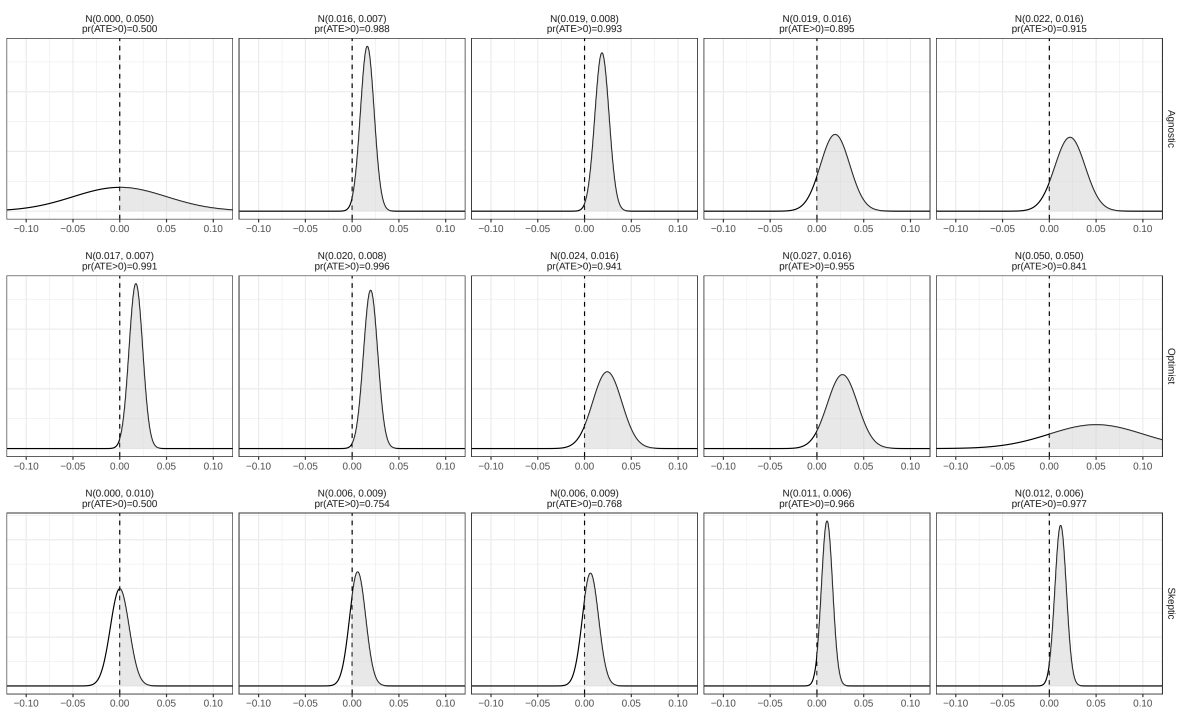

Figure 1: Figure 2 as reproduced by the maintained rewrite.

8 Maintained rewrite verification

Table 7: Maintained rewrite verification: published value against rewrite output.

Location	Exp.	Arm	Model	Quantity	Paper	Rewrite	Match
table_2	all	Control	n	N	16	16	1
table_2	all	Adjacent	n	N	49	49	1
table_2	all	Treated	n	N	23	23	1
table_2	all	Control	n	N	13	13	1
table_2	all	Adjacent	n	N	41	41	1
table_2	all	Treated	n	N	15	15	1
table_2	all	Control	n	N	25	25	1
table_2	all	Adjacent	n	N	76	76	1
table_2	all	Treated	n	N	30	30	1
table_2	all	Control	n	N	24	24	1
table_2	all	Adjacent	n	N	44	44	1
table_2	all	Treated	n	N	20	20	1
table_3	1	Treated	M1	coef	0.025	0.0251849231150561	1
table_3	1	Treated	M1	SE	0.027	0.0271953497940833	1
table_3	1	Adjacent	M1	coef	0.037	0.0365245556057411	1
table_3	1	Adjacent	M1	SE	0.027	0.0273484964987663	1
table_3	1	Treated	M2	coef	0.025	0.0246700898107479	1
table_3	1	Treated	M2	SE	0.017	0.0170438046883578	1
table_3	1	Adjacent	M2	coef	0.018	0.0176924658008986	1
table_3	1	Adjacent	M2	SE	0.016	0.0156295800079384	1
table_3	1	all	M1	N	88	88	1
table_3	1	all	M1	R2	0.031	0.0314509025405569	1
table_3	1	all	M2	R2	0.823	0.822500240891992	1
table_4	2	Treated	M1	coef	0.009	0.00850373305360868	1
table_4	2	Treated	M1	SE	0.054	0.0539779334852138	1
table_4	2	Adjacent	M1	coef	0.012	0.0117388512564832	1
table_4	2	Adjacent	M1	SE	0.046	0.0463818505970431	1
table_4	2	Treated	M2	coef	-0.014	-0.0142778009663527	1
table_4	2	Treated	M2	SE	0.057	0.0574424251287565	1
table_4	2	Adjacent	M2	coef	0.004	0.00428044547072836	1
table_4	2	Adjacent	M2	SE	0.045	0.0453949744301064	1
table_4	2	all	M1	N	69	69	1
table_4	2	all	M1	R2	0.001	0.00108065739965002	1
table_5	3	Treated	M1	coef	0.042	0.0415499685697058	1
table_5	3	Treated	M1	SE	0.016	0.0159171166363333	1
table_5	3	Adjacent	M1	coef	0.042	0.042377169733312	1
table_5	3	Adjacent	M1	SE	0.013	0.0132163225089229	1
table_5	3	Treated	M2	coef	0.018	0.0183687447195372	1
table_5	3	Treated	M2	SE	0.009	0.00860843709003946	1
table_5	3	Adjacent	M2	coef	0.018	0.0184661680108188	1
table_5	3	Adjacent	M2	SE	0.007	0.0065603412388164	1
table_5	3	all	M1	N	131	131	1
table_5	3	all	M1	R2	0.094	0.0939464783867885	1
table_5	3	all	M2	R2	0.825	0.825105729313222	1
table_6	4	Treated	M1	coef	-0.013	-0.0134254300332048	1
table_6	4	Treated	M1	SE	0.028	0.0276296540671851	1
table_6	4	Adjacent	M1	coef	-0.023	-0.0234889507102779	1
table_6	4	Adjacent	M1	SE	0.022	0.0217799790242458	1
table_6	4	Treated	M2	coef	-0.012	-0.0122155097107169	1
table_6	4	Treated	M2	SE	0.026	0.0257990947698161	1
table_6	4	Adjacent	M2	coef	-0.02	-0.0202615460173702	1
table_6	4	Adjacent	M2	SE	0.021	0.0205207147642803	1
table_6	4	all	M1	N	88	88	1
table_6	4	all	M1	R2	0.012	0.0115011646658062	1
table_6	4	all	M2	R2	0.172	0.172251358226275	1
table_7	all	Treated	pooled	coef_exp1	0.025	0.0246700898107479	1
table_7	all	Treated	pooled	SE_exp1	0.017	0.0170438046883578	1
table_7	all	Treated	pooled	coef_exp2	-0.014	-0.0142778009663527	1
table_7	all	Treated	pooled	SE_exp2	0.057	0.0574424251287565	1
table_7	all	Treated	pooled	coef_exp3	0.018	0.0183687447195372	1
table_7	all	Treated	pooled	SE_exp3	0.009	0.00860843709003946	1
table_7	all	Treated	pooled	coef_exp4	-0.012	-0.0122155097107169	1
table_7	all	Treated	pooled	SE_exp4	0.026	0.0257990947698161	1
table_7	all	Treated	pooled	coef_pooled	0.017	0.0165465347227397	1

Table 7: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Exp.	Arm	Model	Quantity	Paper	Rewrite	Match
table_7	all	Treated	pooled	SE_pooled	0.007	0.00730447545153525	1
table_7	all	Adjacent	pooled	coef_exp1	0.018	0.0176924658008986	1
table_7	all	Adjacent	pooled	SE_exp1	0.016	0.0156295800079384	1
table_7	all	Adjacent	pooled	coef_exp2	0.004	0.00428044547072836	1
table_7	all	Adjacent	pooled	SE_exp2	0.045	0.0453949744301064	1
table_7	all	Adjacent	pooled	coef_exp3	0.018	0.0184661680108188	1
table_7	all	Adjacent	pooled	SE_exp3	0.007	0.0065603412388164	1
table_7	all	Adjacent	pooled	coef_exp4	-0.02	-0.0202615460173702	1
table_7	all	Adjacent	pooled	SE_exp4	0.021	0.0205207147642803	1
table_7	all	Adjacent	pooled	coef_pooled	0.015	0.0150868010853195	1
table_7	all	Adjacent	pooled	SE_pooled	0.006	0.00575541650555256	1
table_8	1	Treated	het	coef	0.031	0.0313323003609761	1
table_8	1	Treated	het	SE	0.017	0.0170677333708408	1
table_8	1	Adjacent	het	coef	0.021	0.0209344338704731	1
table_8	1	Adjacent	het	SE	0.014	0.0144758800003209	1
table_8	1	party_share	het	coef	0.027	0.0266351630335268	1
table_8	1	party_share	het	SE	0.042	0.0415866162620613	1
table_8	1	Treated:party_share	het	coef	0.031	0.0310232649627718	1
table_8	1	Treated:party_share	het	SE	0.03	0.0296668745438136	1
table_8	1	Adjacent:party_share	het	coef	0.011	0.0110440779496757	1
table_8	1	Adjacent:party_share	het	SE	0.025	0.025318596318585	1
table_8	1	all	het	N	88	88	1
table_8	1	all	het	R2	0.829	0.828859947456358	1
table_8	2	Treated	het	coef	-0.009	-0.00939986931205115	1
table_8	2	Treated	het	SE	0.06	0.0598700848190547	1
table_8	2	Adjacent	het	coef	0.011	0.0106083651730772	1
table_8	2	Adjacent	het	SE	0.046	0.0460812684704982	1
table_8	2	party_share	het	coef	-0.02	-0.0204280489102419	1
table_8	2	party_share	het	SE	0.033	0.0327925257658679	1
table_8	2	Treated:party_share	het	coef	0.033	0.0334982428503854	1
table_8	2	Treated:party_share	het	SE	0.063	0.0626600647388735	1
table_8	2	Adjacent:party_share	het	coef	0.053	0.0530270828146152	1
table_8	2	Adjacent:party_share	het	SE	0.033	0.0325277799933964	1
table_8	2	all	het	N	69	69	1
table_8	2	all	het	R2	0.277	0.277201723401003	1
table_8	3	Treated	het	coef	0.02	0.0195718144455067	1
table_8	3	Treated	het	SE	0.009	0.00914339319000478	1
table_8	3	Adjacent	het	coef	0.02	0.0197823488638466	1
table_8	3	Adjacent	het	SE	0.007	0.00723695945014714	1
table_8	3	party_share	het	coef	0.028	0.0277688953304519	1
table_8	3	party_share	het	SE	0.009	0.00868969118027666	1
table_8	3	Treated:party_share	het	coef	0.01	0.00956524433211274	1
table_8	3	Treated:party_share	het	SE	0.008	0.007960757426217	1
table_8	3	Adjacent:party_share	het	coef	0.008	0.0076708613615315	1
table_8	3	Adjacent:party_share	het	SE	0.008	0.0077967575316686	1
table_8	3	all	het	N	131	131	1
table_8	3	all	het	R2	0.829	0.829294153064998	1
table_8	4	Treated	het	coef	-0.014	-0.0138509572033166	1
table_8	4	Treated	het	SE	0.026	0.0264086670442199	1
table_8	4	Adjacent	het	coef	-0.02	-0.0199446168952743	1
table_8	4	Adjacent	het	SE	0.021	0.0211109363574492	1
table_8	4	party_share	het	coef	0.02	0.020485916463216	1
table_8	4	party_share	het	SE	0.021	0.0206558145688769	1
table_8	4	Treated:party_share	het	coef	-0.021	-0.0206562354757083	1
table_8	4	Treated:party_share	het	SE	0.027	0.0266582996174303	1
table_8	4	Adjacent:party_share	het	coef	-0.016	-0.0157750345971309	1
table_8	4	Adjacent:party_share	het	SE	0.02	0.0203120880299069	1
table_8	4	all	het	N	88	88	1
table_8	4	all	het	R2	0.181	0.181181057025949	1
ri_table	1	all	RI	p_margin		0.1573	
ri_table	1	all	RI	p_share	0.22	0.2225	1
ri_table	1	all	RI	p_turnout		0.5261	
ri_table	2	all	RI	p_margin		0.8928	
ri_table	2	all	RI	p_share	0.9	0.8974	1
ri_table	2	all	RI	p_turnout		0.6967	
ri_table	3	all	RI	p_margin		0.2633	
ri_table	3	all	RI	p_share	0.02	0.0208	1

Table 7: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Exp.	Arm	Model	Quantity	Paper	Rewrite	Match
ri_table	3	all	RI	p_turnout		0.3319	
ri_table	4	all	RI	p_margin		0.8773	
ri_table	4	all	RI	p_share	0.77	0.771	1
ri_table	4	all	RI	p_turnout		0.8514	
appendix_C1	1	Treated	M1	coef	15.582	15.5820882919866	1
appendix_C1	1	Treated	M1	SE	28.327	28.3269024547458	1
appendix_C1	1	Adjacent	M1	coef	-9.242	-9.24162075983511	1
appendix_C1	1	Adjacent	M1	SE	27.305	27.3053512451301	1
appendix_C1	1	Treated	M2	coef	34.786	34.7862791180129	1
appendix_C1	1	Treated	M2	SE	18.207	18.2073735421737	1
appendix_C1	1	Adjacent	M2	coef	11.713	11.7130625829899	1
appendix_C1	1	Adjacent	M2	SE	17.885	17.8847907177863	1
appendix_C1	1	Treated	M1_turnout	coef	53.002	53.0017430795609	1
appendix_C1	1	Treated	M1_turnout	SE	48.748	48.7480177941925	1
appendix_C1	1	Adjacent	M1_turnout	coef	110.329	110.329339988732	1
appendix_C1	1	Adjacent	M1_turnout	SE	52.796	52.7959163940458	1
appendix_C1	1	Treated	M2_turnout	coef	10.331	10.3313788764894	1
appendix_C1	1	Treated	M2_turnout	SE	15.741	15.7409806269364	1
appendix_C1	1	Adjacent	M2_turnout	coef	11.914	11.9137583664288	1
appendix_C1	1	Adjacent	M2_turnout	SE	11.759	11.7586007130588	1
footnote_4	all	all	intext	total_turnout	241613	241613	1
footnote_4	all	all	intext	cost_per_vote_direct	3.18	3.175958831588	1
footnote_4	all	all	intext	cost_per_vote_total	1.69	1.68722812928112	1
figure_2	Agnostic	Prior		posterior_mean	0	0	1
figure_2	Agnostic	Prior		posterior_sd	0.05	0.05	1
figure_2	Agnostic	Prior		pr_ate_positive	0.5	0.5	1
figure_2	Agnostic	Update 1		posterior_mean	0.022	0.022102	1
figure_2	Agnostic	Update 1		posterior_sd	0.016	0.016132	1
figure_2	Agnostic	Update 1		pr_ate_positive	0.915	0.914663	1
figure_2	Agnostic	Update 2		posterior_mean	0.019	0.019442	1
figure_2	Agnostic	Update 2		posterior_sd	0.016	0.015531	1
figure_2	Agnostic	Update 2		pr_ate_positive	0.895	0.89468	1
figure_2	Agnostic	Update 3		posterior_mean	0.019	0.018621	1
figure_2	Agnostic	Update 3		posterior_sd	0.008	0.007529	1
figure_2	Agnostic	Update 3		pr_ate_positive	0.993	0.993304	1
figure_2	Agnostic	Update 4		posterior_mean	0.016	0.016201	1
figure_2	Agnostic	Update 4		posterior_sd	0.007	0.007228	1
figure_2	Agnostic	Update 4		pr_ate_positive	0.988	0.987502	1
figure_2	Optimist	Prior		posterior_mean	0.05	0.05	1
figure_2	Optimist	Prior		posterior_sd	0.05	0.05	1
figure_2	Optimist	Prior		pr_ate_positive	0.841	0.841345	1
figure_2	Optimist	Update 1		posterior_mean	0.027	0.027307	1
figure_2	Optimist	Update 1		posterior_sd	0.016	0.016132	1
figure_2	Optimist	Update 1		pr_ate_positive	0.955	0.954743	1
figure_2	Optimist	Update 2		posterior_mean	0.024	0.024267	1
figure_2	Optimist	Update 2		posterior_sd	0.016	0.015531	1
figure_2	Optimist	Update 2		pr_ate_positive	0.941	0.940907	1
figure_2	Optimist	Update 3		posterior_mean	0.02	0.019755	1
figure_2	Optimist	Update 3		posterior_sd	0.008	0.007529	1
figure_2	Optimist	Update 3		pr_ate_positive	0.996	0.995651	1
figure_2	Optimist	Update 4		posterior_mean	0.017	0.017246	1
figure_2	Optimist	Update 4		posterior_sd	0.007	0.007228	1
figure_2	Optimist	Update 4		pr_ate_positive	0.991	0.991484	1
figure_2	Skeptic	Prior		posterior_mean	0	0	1
figure_2	Skeptic	Prior		posterior_sd	0.01	0.01	1
figure_2	Skeptic	Prior		pr_ate_positive	0.5	0.5	1
figure_2	Skeptic	Update 1		posterior_mean	0.006	0.006318	1
figure_2	Skeptic	Update 1		posterior_sd	0.009	0.008625	1
figure_2	Skeptic	Update 1		pr_ate_positive	0.768	0.768064	1
figure_2	Skeptic	Update 2		posterior_mean	0.006	0.005864	1
figure_2	Skeptic	Update 2		posterior_sd	0.009	0.008529	1
figure_2	Skeptic	Update 2		pr_ate_positive	0.754	0.754103	1
figure_2	Skeptic	Update 3		posterior_mean	0.012	0.012059	1
figure_2	Skeptic	Update 3		posterior_sd	0.006	0.006059	1
figure_2	Skeptic	Update 3		pr_ate_positive	0.977	0.976715	1
figure_2	Skeptic	Update 4		posterior_mean	0.011	0.01079	1
figure_2	Skeptic	Update 4		posterior_sd	0.006	0.005898	1
figure_2	Skeptic	Update 4		pr_ate_positive	0.966	0.966317	1

Table 7: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Exp.	Arm	Model	Quantity	Paper	Rewrite	Match

Every row with a published counterpart carries `match_rewrite = 1`: **191** values produced by the maintained rewrite, all matching the published paper to reported precision. The remaining 8 are the margin and turnout randomization inference p-values, which the article does not state.

9 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-01

Table 9: Package versions used for the run behind this report.

Package	Version
estimatr	1.0.6
metafor	5.0.1
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
here	1.0.2
gridExtra	2.3